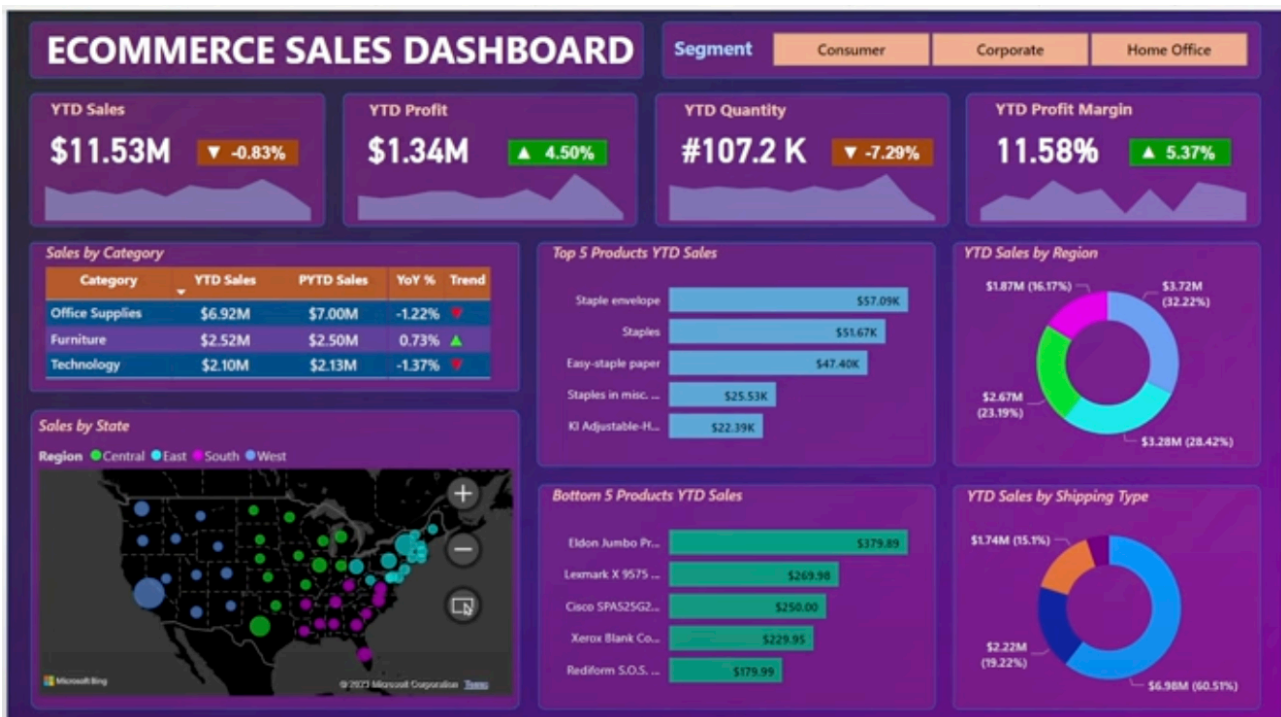


E Commerce Sales Dashboard

Working :

- right top slicer : consumer , corporate , home office and dashboard changes according to them.
- All the values KPIs are dynamic here and it is dynamic in such a way that if it is increasing the background colour will change into green and if it is decreasing the background colour will change to red which will give us the indication that its inc or dec.
- In sales by category I made interactive filter for seeing office supplies or furniture or technology.
- Similarly if we want to see something by region we can see all the values will change with respect to region.
- We can also see by sales.
- It will give an enormous number of insights to our customer , stakeholders to drive there business.
- At top there is a KPI banner which Shows : year to date sales , profit , quantity , profit margin and there is a trend line in each col at bottom and whenever mouse pointer goes to that trend line it shows the sum of sales.
- With respect to segment row everything is dynamic.
- Sales by sales is a bubble chart for particular geographical location .. and its for US ... the bubble size implies the number of sales in that particular state .. and categories with respect to region.
- Then we see 4 boxes showing year to year top 5 products , sales by region , bottom 5 products , shipping type.
- We will use mssql server connection.
- Data is taken from kaggle website .. then cleaned that data



Steps in project :

- problem statement
- Import data in ms sql server
- Connecting power bi to ms sql db
- Data cleaning
- Data processing
- Data modelling : will connect 3 tables in bower bi and will se relation btw them
- Creating data table
- Data visualisation
- Creating dashboard
- Generating insights

Problem Statement :

A US Based E-commerce Sales Company and they want me to create a Sales Dashboard showing information of year to date sale they want to generate insights for the below given scenarios :

- KPI banner showing YTD sales , profit, quantity sold, profit margin.
- On year growth ? , show sparkling or monthly trend to understand the monthly trend each fact.
- YTD sales , PYTD sales , YOY Sales growth for diff customer category and add trend icon to each category.
- YTD sales performance by each state
- Top5 and bottom5 products by sales
- YTD Best and worst performing region all over country
- YTD sales by shipping type to get the best shipping type percentage

Learned from this project :

- how to connect powerBI to MS SQL server and flat files
- Data modelling
- Data cleaning in Power Query
- How to create a Date Table in Power BI
- Time Intelligence function (TOTALYTD, SAMEPERIODLASTYEAR, etc)
- Creating Dynamic and Complex KPI's
- Basic to Advanced Dax Queries
- Conditional Formatting's, Adding dynamic icons in Power BI ... colours changing.
- Different DAX functions like Calculate, Sum, Sumx, Filter, values, selectedvalue, return, concatenate, divide, var, etc
- Creating different charts, maps and formatting then
- Generating insights from charts
- Export report databases.

Data :

- This is all dummy data made for visualisation purpose.
- 21 fields : customer id, name - first last , category segment , customer city , customer state , country , delivery status , profit per segment , shipping type , days for shipment , ship date , item discount , sales per order etc.
- 113k rows
- now in second file it give info about US states longitude and latitude codes... jaha customer order kr rha .
- We will join together both sheets and then start analysing.
- Study ur data first always.

Next step : upload this data to ms sql server

- now data file is .csv file ...
- Now opened SQL Server Management Studio : created a new database named Ecommerce_Database and us_state_long_lat_codes then added the flat files i.e. both the files .. and changed the nvarchar to varchar ..wherever string datatype I am changing it to varchar

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded, showing 'Ecommerce_Database' and its tables. The 'Tables' folder is expanded, showing 'dbo.us_state_long_lat_codes'. In the center, a query window displays the command: `select * from ecommerce_data`. Below the query window, the 'Results' pane shows a table with 17 rows of data. The columns are: customer_id, customer_first_name, customer_last_name, category_name, product_name, customer_segment, customer_city, customer_state, customer_country, customer_region, delivery_status, and order_date.

customer_id	customer_first_name	customer_last_name	category_name	product_name	customer_segment	customer_city	customer_state	customer_country	customer_region	delivery_status	order_date
1	Mary	Fuller	Office Supplies	Xerox 1913	Corporate	New Rochelle	New York	United States	East	Shipping on time	2022-05-1
2	Alan	Edelman	Office Supplies	#5 3/4 Gummed Flap White Envelopes	Corporate	Houston	Texas	United States	Central	Shipping canceled	2022-06-2
3	Mary	Geyman	Office Supplies	Bekins 8 Outlet Surge Protector	Consumer	Louisville	Kentucky	United States	South	Late delivery	2022-06-5
4	Raymond	Easton	Office Supplies	GBC VeldBinder Manual Binding System	Corporate	Chicago	Illinois	United States	Central	Late delivery	2022-06-1
5	Mary	Gonzalez	Furniture	Eldon Pizazz Desk Accessories	Home Office	Philadelphia	Pennsylvania	United States	East	Late delivery	2022-05-4
6	Joe	Leatherbury	Office Supplies	Hoover Shoulder Vac Commercial Portable Vacuum	Home Office	Columbia	South Carolina	United States	South	Shipping on time	2022-06-2
7	Mary	Etezadi	Office Supplies	Rogers Handheld Barrel Pencil Sharpener	Home Office	Henderson	Kentucky	United States	South	Advance shipping	2022-10-4
8	Andrea	O'Briant	Office Supplies	Dixon Prang Watercolor Pencils, 10-Color Set with B...	Corporate	Athens	Georgia	United States	South	Shipping on time	2022-01-2
9	Mary	Etezadi	Office Supplies	Ibico Hi-Tech Manual Binding System	Home Office	Henderson	Kentucky	United States	South	Shipping on time	2022-03-4
10	Mary	Etezadi	Office Supplies	Alliance Super-Size Bands, Assorted Size	Home Office	Henderson	Kentucky	United States	South	Late delivery	2022-03-1
11	Mary	Etezadi	Office Supplies	Southworth 25% Cotton Granite Paper & Envelopes	Home Office	Henderson	Kentucky	United States	South	Advance shipping	2022-11-4
12	Richard	Pak	Technology	Logitech K350 2.4Ghz Wireless Keyboard	Consumer	Madison	Wisconsin	United States	Central	Late delivery	2022-06-1
13	Sarah	Wiener	Furniture	DAX Value U-Channel Document Frames, Easel Back	Consumer	Seattle	Washington	United States	West	Late delivery	2022-05-1
14	Eugene	Pryor	Technology	PowerGen Dual USB Car Charger	Home Office	Yonkers	New York	United States	East	Late delivery	2022-04-3
15	Mary	Ludtke	Office Supplies	Sensible Storage WireTech Storage Systems	Consumer	Yonkers	New York	United States	East	Advance shipping	2022-05-2
16	David	Sundaresam	Office Supplies	Acco Four Pocket Poly Ring Binder with Label Hold...	Consumer	Huntville	Texas	United States	Central	Shipping on time	2022-02-4
17	Mary	Seite	Office Supplies	Newell 312	Consumer	Laredo	Texas	United States	Central	Late delivery	2021-02-5

Fig : showing databases loaded in SQL server Management Studio

Next step : connect powerBI to above flat files.

- Opened a new powerBI worksheet and then imported data from SQL server then loaded them (both).
- Now joining both the tables via customer_state in Ecommerce_Database n name in us_state_long_lat_codes ... in a one to many relationship

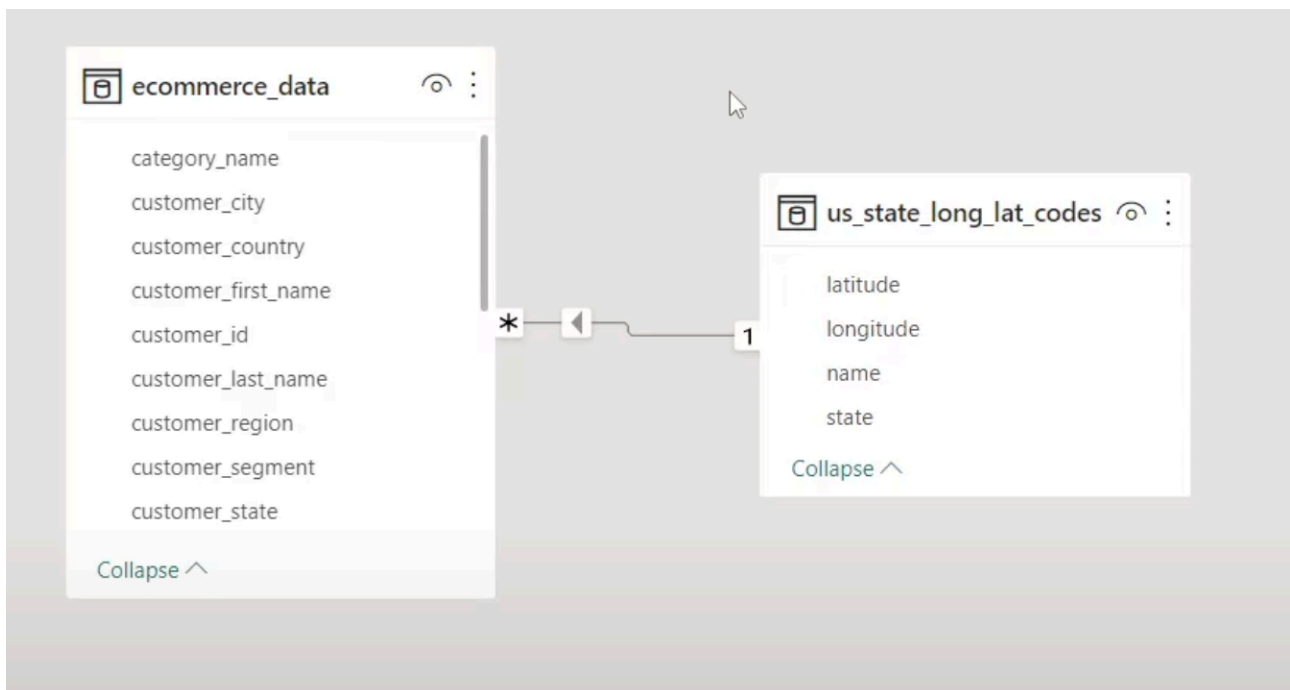


Fig : showing one to many relationship btw tables (customer_state and name)

Now what else I could do, if I do not wanna use SQL Server Management Studio, is that directly from my powerBI load the .csv file

Now re-checking data in powerBI to check if its correctly added or not ...and checked order_date data-type similarly checked whole numbers , decimal numbers , text , etc

Now designing KPIs ... but first creating a new date table which will to use time division functions

- naming date table as calendar ...I have 2 years of data....

```
1 Calendar = CALENDAR(MIN(ecommerce_data[order_date]), TODAY())
```

Fig: date table named as Calendar

- now instead of using TODAY() I am using max date as I have history data :

```
1 Calendar = CALENDAR(MIN(ecommerce_data[order_date]), MAX(ecommerce_data[order_date]))
```

Fig : showing Calendar table

- now created a new column named year:

```
1 Year = YEAR('Calendar'[Date])
```

- now last col for month :

```
1 Month = FORMAT('Calendar'[Date], "mmm")
```

Now connecting order_date to date (in Calendar table) with a cardinality of One to Many

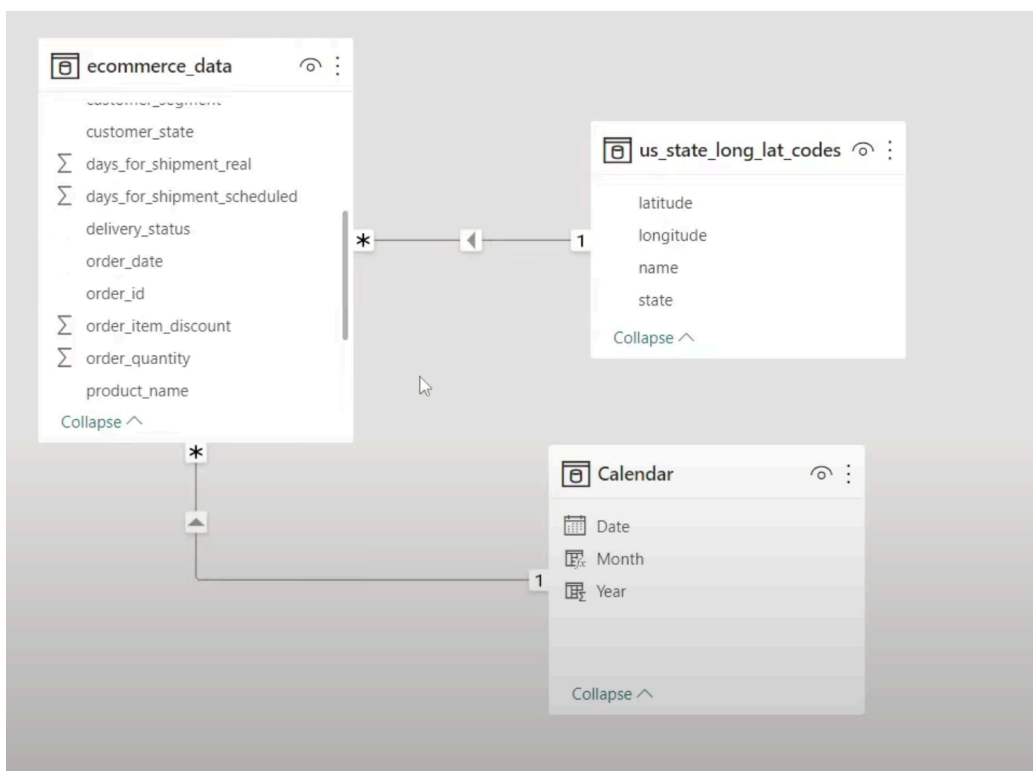


Fig : showing One to Many connection of Table ecommerce_data and Calendar

Now Before Starting , putting a background image by going to format your report page to then canvas background then browsed the image then adjusted the transparency to 0% .

Now adding shapes with colour code : #8D248D and given title of this dashboard (54 x 614)
Then similarly (133 x 295) for the first YTD sale ..I have selected Card then designing KPI ... using time intelligence function to determine the YTD sale for that new measure(below shown).

```
1 YTD Sales = TOTALYTD(SUM(ecommerce_data[sales_per_order]), 'Calendar'[Date])
```

then turned off the effect and background then formatted it to \$ as its US data.

Now adding Area chart and I need chart to be sorted by month by month(jan to dec) in order to do that firstly making a new col in calendar table to name Month Number :

```
1 Month Number = MONTH('Calendar'[Date])
```

This will give the month number, then removed everything and the area chart is a graph btw months vs sum of sales per order for YTD Sales, and additionally only for this table I have created a filter for year 2022 data for the above graph.

Now I will make a on year sales growth which is dynamic i.e. according to the sales change colour from green to red show dec In sales.

But Firstly created a new measure named : PYTD Sales , used a calculate function :

```
1 PYTD Sales = CALCULATE(SUM(ecommerce_data[sales_per_order]), DATESYTD(SAMEPERIODLASTYEAR('Calendar'[Date])))
```

Now I have to determine Year on sales (YoY Sale) , created a new measure ,

```
1 YoY Sales = ([YTD Sales]- [PYTD Sales])/ [PYTD Sales]
```

Now converted it into percentage (%) form, now doing some formatting , now adding background shape/ colour , now for trend icon , created a new measure for that named as Sales Icon :
For +ve and -ve , used a unichar function for it ,

```
1 Sales Icon = var postive_icon = UNICHAR(9650)
2               var negative_icon = UNICHAR(9660)
3               var result = IF([YoY Sales]>0, postive_icon, negative_icon)
4               return result
```

This Sales Icon will show upward or downward arrow according to the Year on sales.

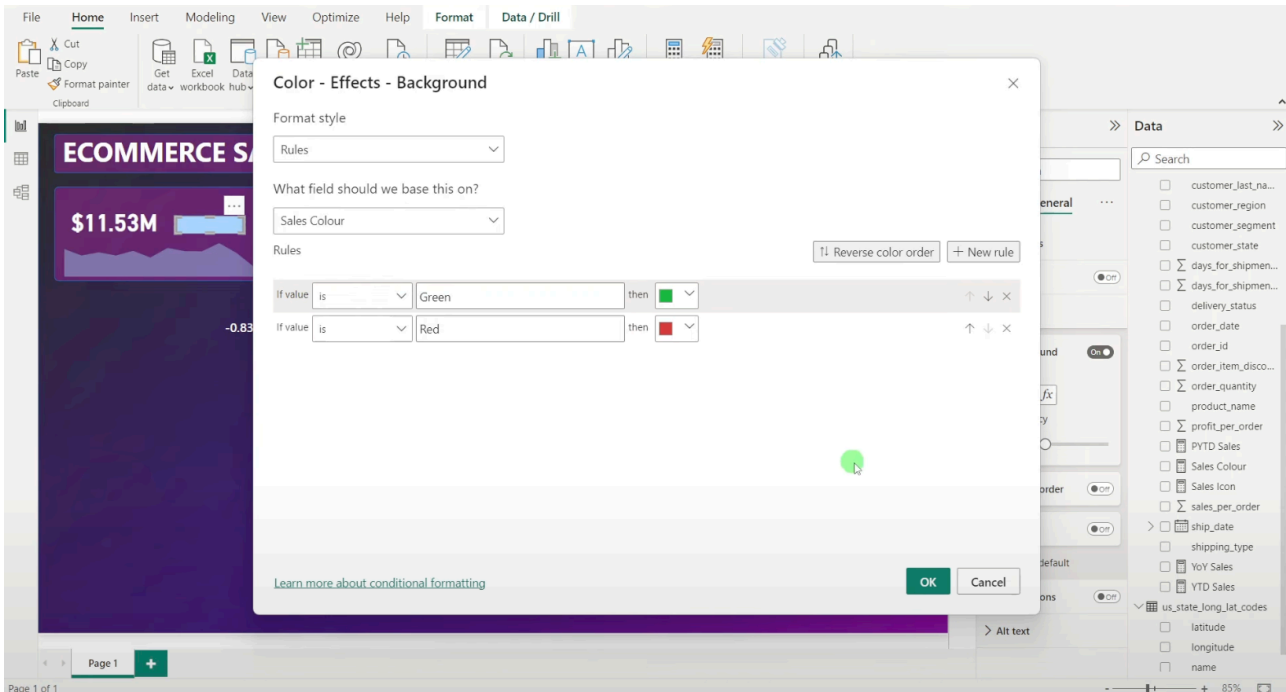
Now last thing about this box is change of colour , for this I have to write again a calculation field ,
New measure as Sales Icon Colour :

```
1 Sales Icon Colour = IF([YoY Sales]>0, "Green", "Red")
```

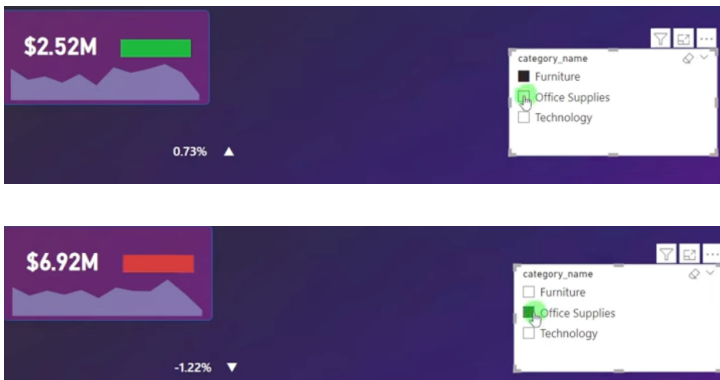
Now dynamic colour for the shape :

```
1 Sales Colour = IF([YoY Sales]>0, "Green", "Red")
```

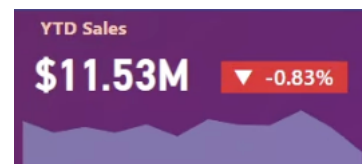
Now :



Now according to category we can also check the YTD Sales :



Final :



Now similarly made all four KPIs :

YTD Profit -

```
1 YTD Profit = TOTALYTD(SUM(ecommerce_data[profit_per_order]), 'Calendar'[Date])
```

Graph : profit_per_order vs date

```
1 PYTD Profit = CALCULATE(SUM(ecommerce_data[profit_per_order]), DATESYTD(SAMEPERIODLASTYEAR('Calendar'[Date])))
```

```

1 YoY Profit = ([YTD Profit]- [PYTD Profit])/ [[PYTD Profit]]
1 Profit Icon = var positive_icon = UNICHAR(9650)
2               var negative_icon = UNICHAR(9660)
3               var result = IF([YoY Profit]>0, positive_icon, negative_icon)
4               return result

```

```

1 Profit Colour = IF([YoY Profit]>0, "Green", "Red")

```

Now YTD Quantity

YTD Quantity = TOTALYTD(SUM(ecommerce_data[order_quantity]), 'Calendar'[Date])

For adding a special character “#” I have to write a new measure :

YTD Concatenated Qty = CONCATENATED("#", FORMAT([YTD Qty]/1000 , "0.0 k"))

PYTD Quantity = CALCULATE(SUM(ecommerce_data[order_quantity]) ,
DATESYTD(SAMEPERIODLASTYEAR('Calendar'[Date])))

YOY Qty = ([YTD Qty] - [PYTD Qty]) / [PYTD Qty]

```

Qty Icon = var positive_icon = UNICHAR(9650)
           var negative_icon = UNICHAR(9660)
           var result = IF([YoY Qty] > 0 , positive_icon , negative_icon)
           return result

```

Qty Colour = IF([YoY Qty] > 0 , "Green" , "Red")

YTD Profit Margin

Profit Margin = SUM(ecommerce_data[profit_per_order]) /
SUM(ecommerce_data[sales_per_order])

YTD Profit Margin = TOTALYTD([Profit Margin] , "Calendar"[Date])

PYTD Profit Margin = CALCULATE([Profit Margin], DATESYTD(SAMEPERIODLASTYEAR('Calendar'[Date])))

YoY Profit Margin = ([YTD Profit Margin] - [PYTD Profit Margin]) / [PYTD Profit Margin]

```

Profit Margin Icon = var positive_icon = UNICHAR(9650)
                    var negative_icon = UNICHAR(9660)
                    var result = IF([YoY Profit Margin] > 0 , positive_icon , negative_icon)
                    return result

```

Profit Margin Colour = IF([YoY Profit margin] > 0 , "Green" , "Red")

Now

Next chart is **Sales by Category Chart** :

Taking a Matrix's Col as : category_name , YTD Sales , PYTD Sales , YoY Sales

Then named sales icon as trend

Some insights : for business to check by category of the sale of products and from past year sales are going up or down and in which month sales are going up or down or max or min and acc to which they can bring some offers for mini sales month

Next chart is **Sales by State** :

Will be using us_state_lat_long table and map chart , putting longitude and latitude, bubble size = YTD Sales , giving it region by region colour , now giving diff colour for North , south , west , east .

Next chart is Bar Chart for **Top 5 Products by YTD Sales**

Using bar chart for YTD Sales of Top 5 Products

Next chart is Bar Chart for **Bottom 5 Products by YTD Sales**

Using bar chart for YTD Sales of Bottom 5 Products

Now **YTD Sales by Region**

Using donut chart for it

Similarly **YTD Sales by Region**

Now last step is **Slicer**

Slicer for customer_segment , using tiled format.

Changing interaction :

For Sales by Category filtering out bar n donut charts ,

Similarly for Sales by State filtering out ... instead of highlights.

Done Bhaiiii.....